

Computational Biofluid Mechanics

Patient-Specific Computational Fluid Dynamics of the Airways

Segmentation, Mesh Generation, and Flow Analysis in Tracheomalacia

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Repository: github.com/henrivanovermeire/tracheomalacia_cfd

Reproducible project report; analyses still in progress are identified explicitly in the text.

Abstract

Tracheomalacia is characterised by excessive collapse or narrowing of the tracheal lumen and can substantially alter respiratory airflow. This project uses patient-specific computed tomography, three-dimensional airway segmentation, and computational fluid dynamics (CFD) to compare airflow before and after intervention. Airway surfaces are segmented in 3D Slicer, clipped along VMTK-derived centerlines, extended in the local centerline direction, and closed with planar caps. Tetrahedral volume meshes are generated with Gmsh and steady incompressible airflow is solved using OpenFOAM. This report documents the reproducible processing pipeline, evaluates mesh quality and mesh independence, and compares clinically relevant flow quantities between the preoperative and postoperative geometries. The preoperative minimum lumen area was 1.813 mm^2 , compared with 5.169 mm^2 at the matched postoperative location, giving an area constriction of 64.9%. In the steady postoperative model the local pressure drop was 30.95 Pa, the resistance was 15.41 Pa/(L min), and 73.2% of outlet flow entered the right lung. A four-level HXT mesh study selected the 0.15 mm mesh because its reported metrics differed by no more than 2.02% from the denser mesh while using 48% fewer cells. A complete-cycle transient calculation is configured; its final phase-resolved results are reported only after solver completion.

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1 Introduction

Tracheomalacia involves reduced structural stability of the tracheal wall, which may cause dynamic narrowing and increased respiratory resistance. While medical imaging describes airway anatomy, CFD can quantify how geometric narrowing influences velocity, pressure loss, flow distribution, and local shear-related quantities.

The objectives of this project are to:

1. construct patient-specific preoperative and postoperative airway geometries from computed tomography;
2. establish a reproducible segmentation-to-simulation workflow;
3. quantify airflow and pressure differences between the two cases; and
4. assess whether the numerical conclusions are independent of mesh resolution.

1.1 Research questions

- How does the intervention change the minimum airway cross-section?
- How does the pressure drop across the trachea change?
- Where do peak velocities and recirculation regions occur?
- How is flow divided among the distal outlets?
- Are the reported quantities sufficiently mesh independent?

2 Critical review of a selected CFD paper

The selected paper is Taherian et al.'s patient-specific CFD evaluation of excessive dynamic airway collapse before and after tracheal stenting [1]. Each of the five required subsections below is shorter than 150 words. No paper figures are reproduced, remaining within the permitted maximum of three.

2.1 Goal

The study investigated whether patient-specific computational fluid dynamics could quantify the functional consequences of excessive dynamic airway collapse and assess improvement after tracheal stenting. The authors specifically compared inspiratory and expiratory airflow before and after intervention using CT-derived airway geometries. They aimed to determine how stenting affected pressure loss, velocity, wall shear stress, and turbulence, and whether these CFD quantities revealed clinically relevant changes that conventional pulmonary function tests might miss. The underlying question was therefore not merely whether the airway lumen became larger, but whether the anatomical intervention produced a meaningful improvement in respiratory flow behaviour.

2.2 Methods

Four patient-specific airway models represented inspiration and expiration before and after stenting. CT images were segmented in Mimics and simulated in STAR-CCM+. Each rigid model included the trachea and approximately six to eight airway generations. The authors simulated 2.5 s inspiration and 2.5 s expiration with sinusoidal resting-breathing flow. Patient-specific outlet pressure functions were derived in a preliminary step from lobar volume changes,

while the inlet was atmospheric with 10% turbulence intensity. Unsteady RANS with a low-Reynolds-number $k-\omega$ model and second-order spatial and temporal schemes was used. Inlet and outlet extensions measured one and four hydraulic diameters, respectively. Polyhedral meshes with prism layers were tested at 0.9, 1.7, and 3.2 million cells; 1.7 million cells was selected from pressure, velocity, and wall-shear-stress sensitivity.

2.3 Results

Stenting produced the clearest improvement during expiration. The severe pre-stent narrowing generated a high-velocity jet, recirculation, substantial pressure loss, and elevated wall shear stress. The reported pre-/post-stent expiratory pressure difference was approximately 26 cmH₂O, while mean tracheal wall shear stress decreased from 3.98 to 0.793 Pa, an approximately 80% reduction. Velocity variations also became smaller after stenting. Inspiratory changes were comparatively minor; mean wall shear stress increased from 0.0354 to 0.0813 Pa, and the stent introduced some local flow disturbance. Turbulence intensity increased in parts of both phases after intervention. Spirometry changed only modestly—FEV₁ increased by 290 mL—whereas CFD exposed pronounced expiratory functional improvement that pulmonary function testing did not clearly reflect.

2.4 Limitations

The study considered only one patient, so its quantitative findings cannot be generalised to all patients with excessive dynamic airway collapse. The airway walls were rigid even though wall deformation is the defining feature of the disease. Separate end-inspiration and end-expiration geometries represented the extremes of breathing rather than continuously moving anatomy; fluid–structure interaction was omitted because patient-specific tissue properties were unavailable. The upper airway and deeper peripheral generations were truncated, although both may influence tracheal flow and outlet impedance. Outlet pressures were inferred indirectly from lobar volume changes, and local CFD predictions were not validated against invasive pressure or velocity measurements. The RANS turbulence model and selected boundary conditions add model-form uncertainty. Finally, mesh sensitivity was demonstrated primarily for selected quantities in one pre-stent expiratory geometry.

2.5 Evaluation

For this patient, the authors largely answered their research question. The paired pre-/post-stent and inspiration/expiration design isolated the phase in which airway collapse mattered most, and pressure, velocity, and wall shear stress changed coherently after intervention. The contrast with relatively insensitive spirometry supports the claim that CFD can provide additional functional information. The mesh test and patient-informed outlet conditions strengthen the internal comparison. However, the study establishes a patient-specific proof of concept rather than a validated diagnostic or prognostic method. Rigid endpoint geometries cannot reproduce dynamic coupling between airflow and collapse, and a single case cannot establish clinical accuracy or treatment thresholds. Thus, the conclusion that stenting improved this patient’s expiratory airflow is supported, whereas broader claims about CFD-guided management require validation in more patients with dynamic or fluid–structure-interaction models.

3 Imaging data and anatomical cases

Two computed-tomography datasets are analysed: a preoperative case and a postoperative case. Table 1 should be completed from the DICOM metadata and segmentation logs.

Table 1: Imaging and anatomical case information.

Case	Voxel dimensions	Voxel spacing (mm)	Clinical state
Preoperative	[insert]	[insert]	Before intervention
Postoperative	[insert]	[insert]	After intervention

All patient-derived data must be handled according to the applicable privacy, ethics, and data-management requirements. No directly identifying information should appear in this report or repository.

4 Airway segmentation and surface preparation

4.1 Segmentation

The CT series is loaded into 3D Slicer. A case-specific seed is placed in the airway lumen, and connected region growth is performed within a selected Hounsfield-unit interval. Thresholds and seed coordinates are stored under `segmentation/assets/<case>/`. The result is inspected in axial, coronal, sagittal, and three-dimensional views to exclude lung parenchyma and external air while retaining the clinically relevant branches.

Record the final settings in Table 2.

Table 2: Final segmentation settings.

Case	Lower HU	Upper HU	Segmented voxels
Preoperative	[insert]	[insert]	[insert]
Postoperative	[insert]	[insert]	[insert]

4.2 Centerline extraction and clipping

A centerline network is calculated using VMTK. Visually refined clipping points are loaded as the Slicer markup node `AirwayCutEndpoints`. The airway is clipped using the same *Clip Vessel* logic as the SlicerVMTK user interface, with `AirwayNetworkModel` providing the centerline direction.

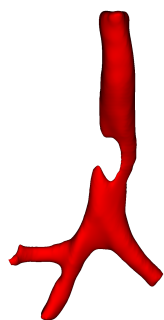
Flow extensions are aligned with the local centerline and adapt to the airway radius at each clipping location. Planar caps close the extensions for CFD. The final Slicer model is `AirwayExtendedSurfaceCapped` and is exported as `meshes/<case>/airways.stl`, preventing the two anatomies from silently overwriting one another.

4.3 Anatomical measurements and degree of constriction

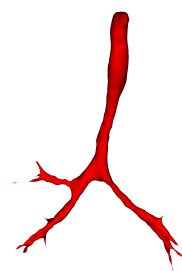
The preoperative constriction is bounded by proximal and distal landmarks on the tracheal centerline. Its length is measured along that centerline. Cross-sections normal to the local centerline are sampled through the narrowed region to locate the minimum lumen area. Because the lumen need not be circular, an equivalent circular diameter is calculated from section area A as

$$D_{\text{eq}} = 2\sqrt{\frac{A}{\pi}}. \quad (1)$$

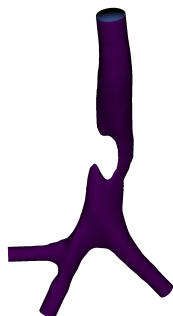
The postoperative area and equivalent diameter are measured using the same procedure at 76.53% of the inlet-to-carina centerline length. The carina is defined from the airway-network bifurcation nearest the airway seed. The primary area-based degree of constriction is



(a) Preoperative cleaned airway segmentation.



(b) Postoperative airway segmentation.



(c) Preoperative clipped, extended, and capped surface.



(d) Postoperative clipped, extended, and capped surface.

Figure 1: Case-specific airway preparation in anatomical frontal view. The top row shows the live Slicer segmentations; the bottom row shows the final CFD surfaces with centerline-directed flow extensions and planar caps.

$$C_A = \left(1 - \frac{A_{\text{pre,min}}}{A_{\text{post,matched}}}\right) \times 100\%. \quad (2)$$

The measured area constriction is 64.9%. For transparency, the corresponding minimum-Feret-diameter reduction is 65.2% and is reported as

$$C_D = \left(1 - \frac{D_{\text{pre,min}}}{D_{\text{post,matched}}}\right) \times 100\%. \quad (3)$$

Table 3: Anatomical measurements obtained from centerline-normal lumen sections in 3D Slicer.

Quantity	Symbol	Value
Preoperative constriction length	L_{pre}	12.157 mm
Preoperative minimum area	$A_{\text{pre,min}}$	1.813 mm ²
Preoperative minimum Feret diameter	$D_{\text{pre,Feret}}$	0.798 mm
Preoperative equivalent diameter	$D_{\text{pre,eq}}$	1.519 mm
Postoperative matched area	$A_{\text{post,matched}}$	5.169 mm ²
Postoperative minimum Feret diameter	$D_{\text{post,Feret}}$	2.292 mm
Postoperative equivalent diameter	$D_{\text{post,eq}}$	2.565 mm
Area constriction	C_A	64.9%
Minimum Feret diameter reduction	$C_{D,\text{Feret}}$	65.2%

The measured dimensions are consistent with the scale reported in paediatric tracheal-surgery literature. Arcieri et al. described 25 children with congenital tracheal stenosis and reported a mean internal diameter of 1.5 mm; the cohort had a median operative age of seven months and median weight of 5.2 kg [2]. This closely matches the 1.519 mm preoperative equivalent diameter obtained here. The smaller minimum Feret diameter of 0.798 mm is not contradictory: it characterises the narrowest caliper dimension of a non-circular lumen, whereas the equivalent diameter is derived from total cross-sectional area. The literature comparison supports dimensional plausibility but is not an independent validation of the patient-specific segmentation or landmark placement.

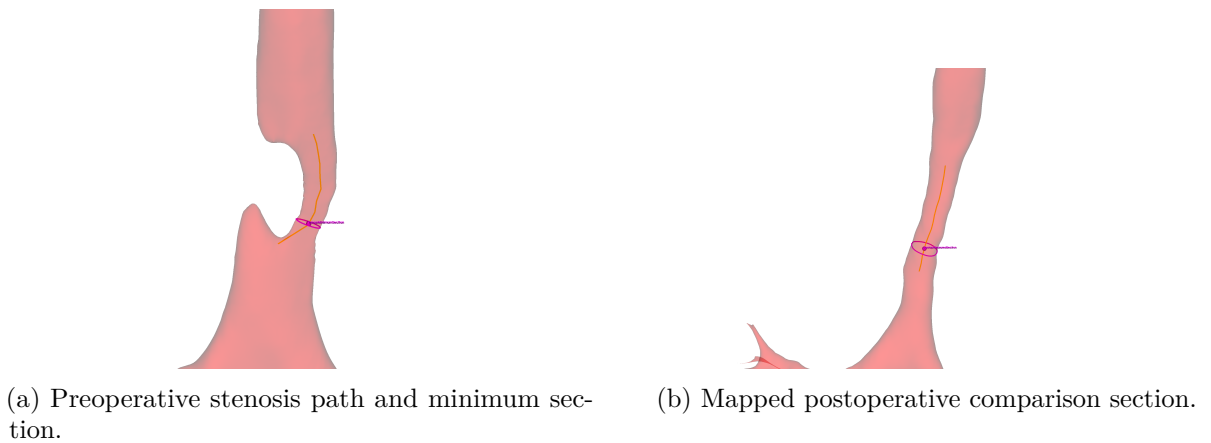


Figure 2: Centerline-based anatomical correspondence. Orange curves show the measured tracheal interval and magenta contours identify the preoperative minimum and corresponding postoperative lumen sections. The postoperative location is mapped using 76.53% of inlet-to-carina centerline distance.

4.4 Qualitative flow behaviour through the constriction

For incompressible flow, continuity $Q = AU$ requires velocity to rise as the area narrows and fall after expansion. Before the stenosis, pressure decreases gradually because viscous wall shear opposes motion; for laminar circular flow, $\Delta p = 128\mu LQ/(\pi D^4)$. Through the stenosis, acceleration converts static pressure into kinetic energy according to Bernoulli's principle, while viscosity and separation add irreversible loss. Downstream, deceleration causes partial pressure recovery, but pressure remains below the upstream trend because separation, mixing, and wall friction dissipate mechanical energy.

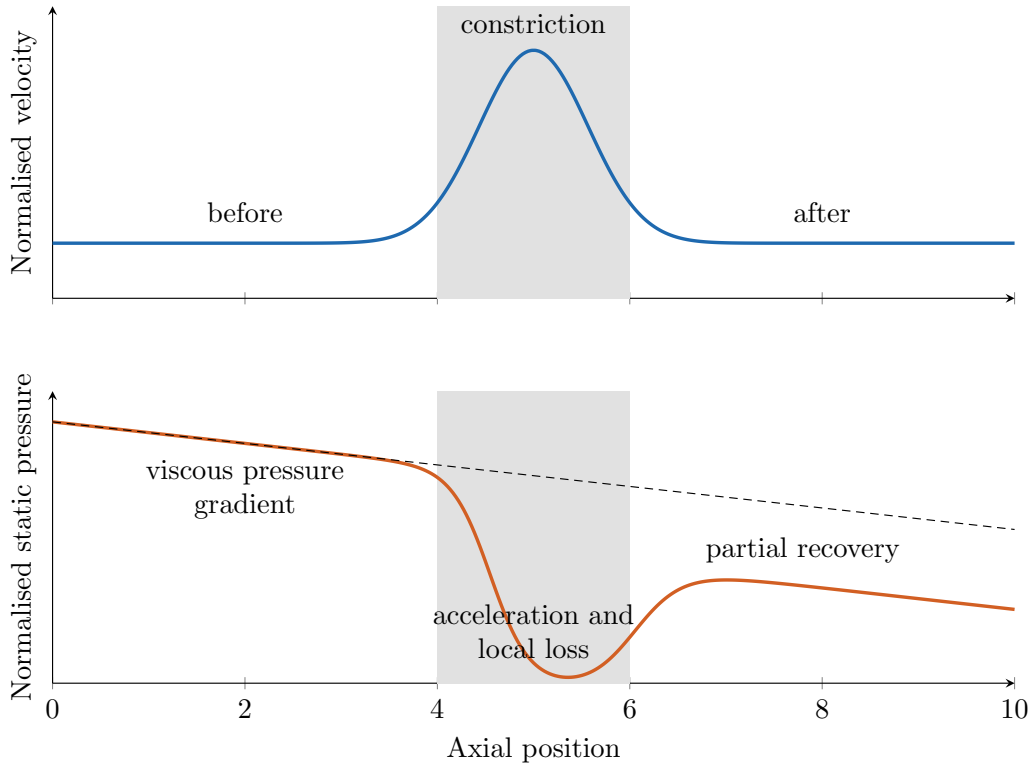


Figure 3: Qualitative velocity and static-pressure evolution through a constriction. Velocity increases as area decreases; pressure falls sharply through the stenosis and recovers only partially downstream. Curves are schematic and not CFD results.

5 Volume mesh generation

The triangulated STL surface is imported into Gmsh. Surface classification and geometry creation produce a closed fluid volume. The physical boundary groups are:

- `inlet`: tracheal inlet;
- `outlet_1`: right superior lobar bronchus;
- `outlet_2`: right inferior lobar bronchus;
- `outlet_3`: left main bronchus;
- `wall`: remaining airway surface; and
- `fluid`: internal volume.

The physical groups are visually verified whenever the STL topology or Gmsh classification changes. The volume consists of first-order tetrahedra with a uniform target characteristic length controlled by `MESH_SIZE`.

Table 4: Replicable mesh settings and quality metrics. All current volume elements are first-order tetrahedra. Add the exact Gmsh version and final values from each `checkMesh` log.

Mesh	Target size (mm)	Points	Cells	Element type	Max. aspect ratio	Max. non-orthogonality
HXT baseline	0.25	39,977	180,163	Tetrahedron	114.14	88.63°
HXT fine 1	0.20	71,198	339,886	Tetrahedron	11.73	80.29°
HXT fine 2	0.15	151,599	776,568	Tetrahedron	12.63	69.82°
HXT densest	0.12	281,330	1,493,440	Tetrahedron	88.00	82.34°

5.1 Near-wall resolution

Boundary-layer prism elements are commonly used near no-slip walls because they resolve steep wall-normal velocity gradients efficiently and with better cell alignment than isotropic tetrahedra. The current Gmsh workflow uses only first-order tetrahedra and does not include dedicated inflation layers. This must not be presented as a hybrid boundary-layer mesh. Its consequences are considered when interpreting wall-adjacent quantities, and adding prism layers is identified as a model improvement.

The Gmsh mesh uses millimetres because it inherits the Slicer coordinate scale. After `gmshToFoam`, all coordinates are multiplied by 10^{-3} to obtain SI units.

6 Mathematical model

6.1 Governing equations

Air is modelled as an incompressible Newtonian fluid. For steady flow, mass and momentum conservation are:

$$\nabla \cdot \mathbf{u} = 0, \quad (4)$$

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad (5)$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and μ is dynamic viscosity. OpenFOAM uses kinematic pressure and the kinematic viscosity $\nu = \mu/\rho$. The configured value is $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$.

The current case uses the laminar model and the steady-state `simpleFoam` solver. The appropriateness of the laminar assumption should be assessed using a Reynolds number based on representative velocity U and diameter D :

$$\text{Re} = \frac{UD}{\nu}. \quad (6)$$

[Calculate Reynolds numbers at the inlet and stenosis and discuss whether a laminar, transitional, or turbulent model is justified.]

6.2 Boundary conditions

A prescribed volumetric flow rate is applied at the tracheal inlet:

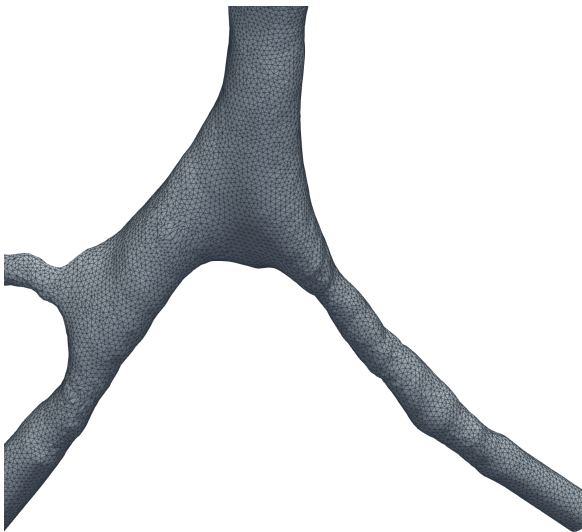
$$Q_{\text{in}} = 3.3333 \times 10^{-5} \text{ m}^3/\text{s}. \quad (7)$$



(a) 0.25 mm HXT baseline; 180,163 volume cells.



(b) Selected 0.15 mm HXT mesh; 776,568 volume cells.



(c) Baseline surface triangulation at the carina.



(d) Selected-mesh surface triangulation at the carina.

Figure 4: Postoperative HXT mesh refinement in matched anatomical frontal views. The complete baseline and selected meshes are shown above, with identically framed carina close-ups below. Both levels use the same geometry, physical surface groups, and HXT algorithm; only target characteristic length differs. Surface-edge density illustrates refinement, while quantitative volume-mesh sensitivity is assessed separately in Section 8.4.

This corresponds to approximately 2 L/min, representative of the stated minute volume for a 10-month-old infant. A no-slip condition is applied at the airway wall. The three distal outlets use the same zero gauge pressure with pressure-compatible velocity conditions. No outlet flow fraction is prescribed; consequently, the flow division is an output of the CFD solution. For a common downstream pressure, each pathway behaves approximately as

$$Q_i \sim \frac{\Delta P}{R_i}, \quad (8)$$

where differences in effective resistance R_i arise from features such as branch diameter, length, branching angle, curvature, and flow separation. Table 5 summarises the implementation.

Table 5: OpenFOAM boundary conditions.

Patch	Velocity $\mathbf{u}$	Pressure p
inlet	flowRateInletVelocity	zeroGradient
outlet_1-3	pressureInletOutletVelocity	fixedValue 0
wall	noSlip	zeroGradient

7 Numerical method

The teaching staff approved the use of 3D Slicer and SlicerVMTK for geometry preparation, Gmsh for meshing, OpenFOAM for CFD, and independent local/remote compute infrastructure. This changes the software implementation but not the required physical analyses or validation.

Table 6: Software and provenance information. Replace placeholders with the immutable versions used for the final simulations.

Stage	Software	Version or identifier
Segmentation	3D Slicer + SlicerVMTK	5.12.2; extension [insert]
Surface/volume meshing	Gmsh	[insert]
CFD	OpenFOAM Docker image	2512; image ID [insert]
Post-processing	ParaView	[insert]
Source state	Git	commit [insert]

7.1 Steady solver

The finite-volume discretisation is implemented in OpenFOAM 2512. Pressure is solved using GAMG and velocity using a symmetric Gauss–Seidel smoothed solver. The SIMPLE algorithm uses two non-orthogonal correctors. The configured residual controls are 10^{-5} for pressure and 10^{-6} for velocity.

The parallel mesh is decomposed using the Scotch method. Simulations are run in the `opencfd/openfoam-default:latest` Docker image. The exact image ID and Git commit are recorded in Table 6.

7.2 Transient solver and breathing waveform

The initially selected 0.15 mm HXT mesh proved computationally impractical for the complete postoperative cycle. The final transient analysis is therefore an explicitly labelled proof of concept on the existing 0.25 mm HXT mesh (180,163 tetrahedra). The incompressible, Newtonian

equations are solved with `pimpleFoam`. A symmetric sinusoidal volumetric-flow boundary condition is prescribed at the tracheal inlet,

$$Q(t) = Q_{\max} \sin\left(\frac{2\pi t}{T}\right), \quad (9)$$

where $T = 2.0$ s and $Q_{\max} = 1.0472 \times 10^{-4}$ m³/s (6.283 L/min). The positive half-cycle represents inspiration and the negative half-cycle expiration. Integrating either half-cycle gives a tidal volume of 66.7 mL, consistent with the assumed 2 L/min minute ventilation at 30 breaths per minute.

The inlet uses `flowRateInletVelocity`. All three outlets have zero fixed kinematic pressure and `pressureInletOutletVelocity`, which permits flow reversal during expiration. The rigid airway wall remains no-slip. Time is discretised with the second-order backward scheme and bounded first-order upwind convection. Three PIMPLE outer correctors, two pressure correctors, and two non-orthogonal correctors are used. An initial time step of 1×10^{-5} s is adjusted subject to $Co_{\max} = 5$ and a maximum step of 2×10^{-4} s; fields are written every 0.02 s. A conservative timing run at $Co_{\max} = 2$ established stable coupling, but projected runtime remained prohibitive. The final $Co_{\max} = 5$ case completed on eight MPI ranks in 65 095 s wall time. This elevated Courant limit, coarse mesh, and diffusive convection scheme permit workflow and phase-trend assessment but preclude claims of temporal or spatial independence. Periodicity also cannot be claimed from the single cycle starting from rest.

At the estimated peak mean velocity of approximately 20.3 m/s in the matched section, the Mach number is about 0.06, supporting the incompressibility approximation. The corresponding Reynolds number is approximately 3500, so transitional effects remain a model limitation. A shorter, more realistic mechanically ventilated inspiratory phase would require a larger peak flow for the same tidal volume and could approach the commonly used $Ma = 0.1$ compressibility threshold.

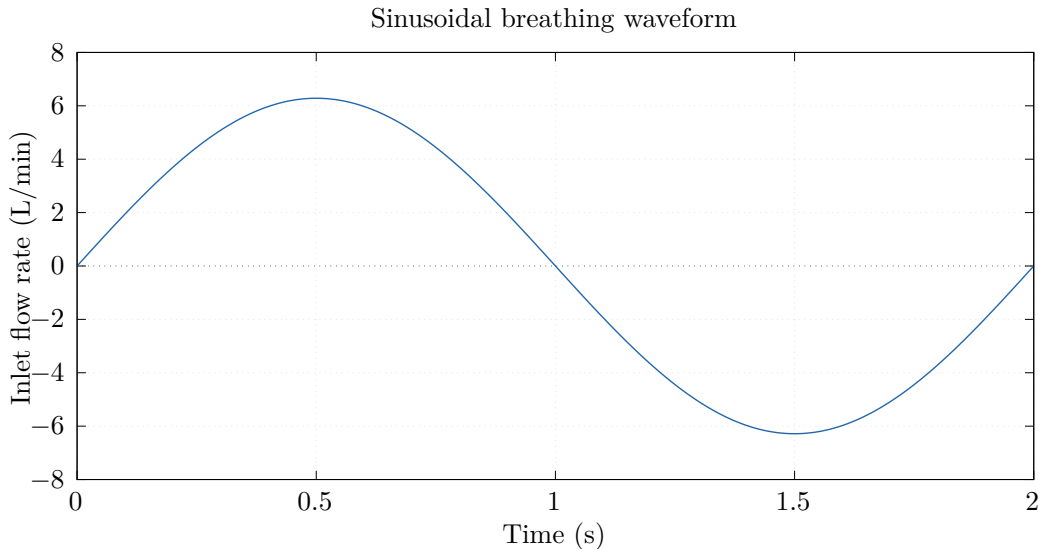


Figure 5: Prescribed sinusoidal inlet flow for the postoperative transient simulation. Positive and negative flow denote inspiration and expiration, respectively.

Table 7: Transient postoperative simulation settings.

Parameter	Value
Solver	<code>pimpleFoam</code>
Breathing-cycle duration	2.0 s
Mesh	HXT, 0.25 mm, 180,163 tetrahedra
Initial / maximum time step	1×10^{-5} s / 2×10^{-4} s
Maximum Courant number	5.0
Field write interval	0.02 s
Simulated cycles	One, starting from rest
PIMPLE correctors	Up to 3 outer, 2 pressure, 2 non-orthogonal
Outer stopping targets	2×10^{-3} (U), 10^{-2} (p)
Linear-equation tolerance	10^{-8} (U and p)
Parallel decomposition	Scotch, 8 MPI ranks

8 Verification

8.1 Mesh quality

Table 8 summarises the principal `checkMesh` metrics for the four postoperative HXT meshes. All meshes contained one connected fluid region. A face was counted as severely non-orthogonal when its non-orthogonality exceeded 70° .

Table 8: Postoperative HXT mesh-quality metrics reported by `checkMesh`. The minimum cell volumes are given after conversion of the geometry from millimetres to metres.

Target size	Max. aspect ratio	Max. non-orthogonality	Severe faces	Max. skewness	Min. volume (m ³)	Regions
0.25 mm	114.14	88.63°	2	1.457	7.90×10^{-15}	1
0.20 mm	11.73	80.29°	2	1.268	3.64×10^{-14}	1
0.15 mm	12.63	69.82°	0	1.196	5.12×10^{-14}	1
0.12 mm	88.00	82.34°	19	1.396	8.71×10^{-16}	1

The 0.15 mm mesh had the strongest overall quality: its maximum non-orthogonality remained below 70° , it contained no severely non-orthogonal faces, and it had the lowest maximum skewness of the series. Its maximum aspect ratio of 12.63 was also substantially lower than those of the 0.25 mm and 0.12 mm meshes. Refining the target size did not improve every quality measure monotonically. In particular, the 0.12 mm mesh introduced 19 severely non-orthogonal faces, a maximum aspect ratio of 88.00, and the smallest cell volume in the series. This local deterioration is consistent with its poorer solver convergence and shows why cell count alone is not a sufficient mesh-selection criterion. Nevertheless, all four meshes remained single connected domains and their maximum skewness was below the OpenFOAM internal-cell warning limit of 4. The quality results therefore support selection of the 0.15 mm mesh for the final analysis.

8.2 Convergence and mass conservation

The SIMPLE solution met residual control after 589 iterations. Final initial residuals were 9.93×10^{-7} , 7.57×10^{-7} , and 6.48×10^{-7} for U_x , U_y , and U_z , respectively, and 4.92×10^{-6} for pressure. These are below the configured criteria of 10^{-6} for velocity and 10^{-5} for pressure. Final outlet-flow integration is still used to verify that

$$Q_{\text{in}} + \sum_{i=1}^3 Q_{\text{out},i} \approx 0, \quad (10)$$

using a consistent outward-normal sign convention. Report the relative mass imbalance.

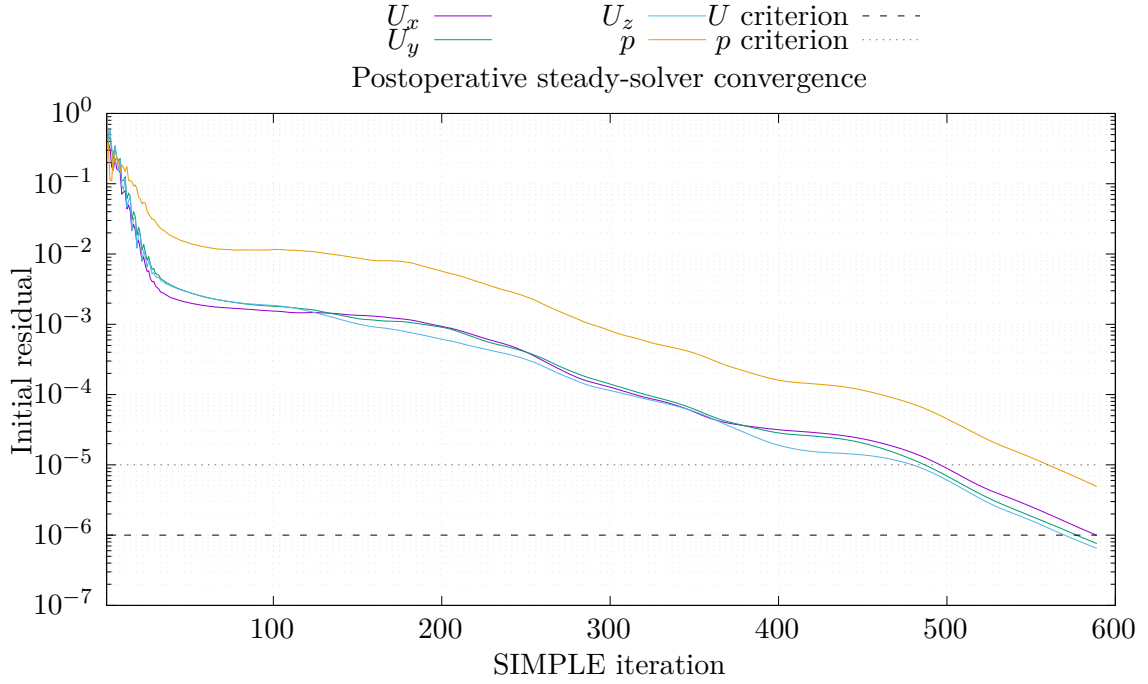


Figure 6: Initial residual evolution for the postoperative steady simulation. Dashed and dotted lines indicate the configured velocity and pressure convergence criteria. The SIMPLE algorithm stopped at iteration 589 after all criteria were satisfied.

8.3 Effect of additional steady iterations

Assignment 4 requests an assessment of whether 1000 additional iterations alter an axial-velocity parameter. A fixed centerline-normal section before the bifurcation is used, and its area-averaged axial velocity is recorded before and after continuation without reinitialising the solution.

Table 9: Sensitivity of the selected axial-velocity monitor to 1000 additional steady iterations.

Parameter	Iteration 2000	Iteration 3000	Relative change (%)
Area-averaged axial velocity	[insert]	[insert]	[insert]

8.4 Mesh-independence assessment

At least three postoperative mesh levels are preferred so that a trend can be assessed rather than only a pairwise difference. Each parameter of interest is plotted against the number of volume elements. For mesh i and the densest mesh d , the Assignment 5 percentage difference is defined as

$$\delta_{\phi,i} = \frac{|\phi_i - \phi_d|}{|\phi_d|} \times 100\%. \quad (11)$$

Table 10: Mesh-independence comparison.

Quantity	0.25 mm	0.20 mm	0.15 mm	0.12 mm
Resistance (Pa/(L/min))	15.153	16.361	17.544	17.906
Peak velocity (m/s)	9.447	9.573	9.670	9.822
Right-lung fraction (%)	73.664	73.619	73.686	73.397
Right-superior share (%)	15.231	15.597	16.017	16.250
Solver iterations	596	770	1611	2000*

*Iteration limit reached without satisfying residual control.

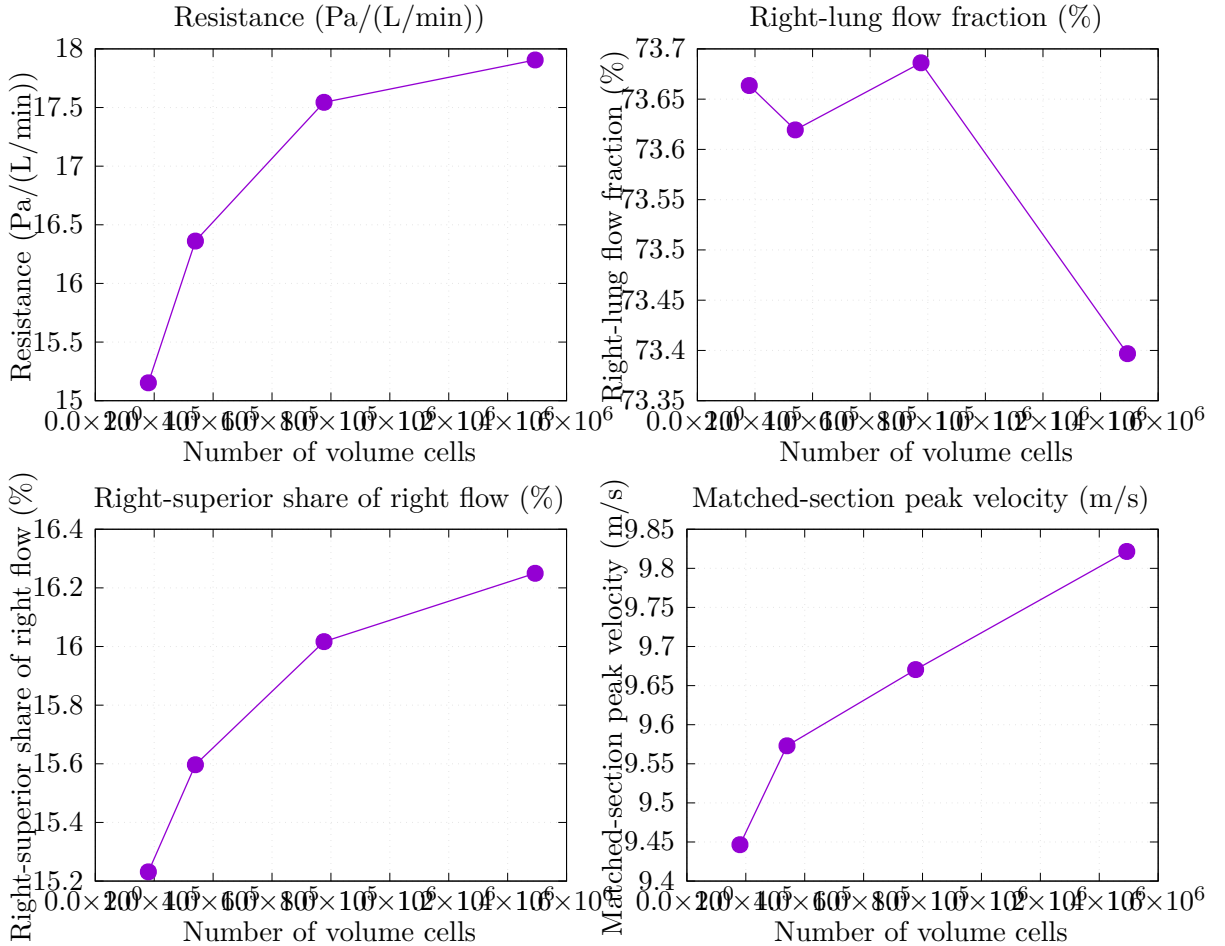


Figure 7: Postoperative mesh-sensitivity metrics as functions of actual volume cell count. All meshes use identical geometry, boundary conditions, sampling planes, and HXT tetrahedral meshing strategy.

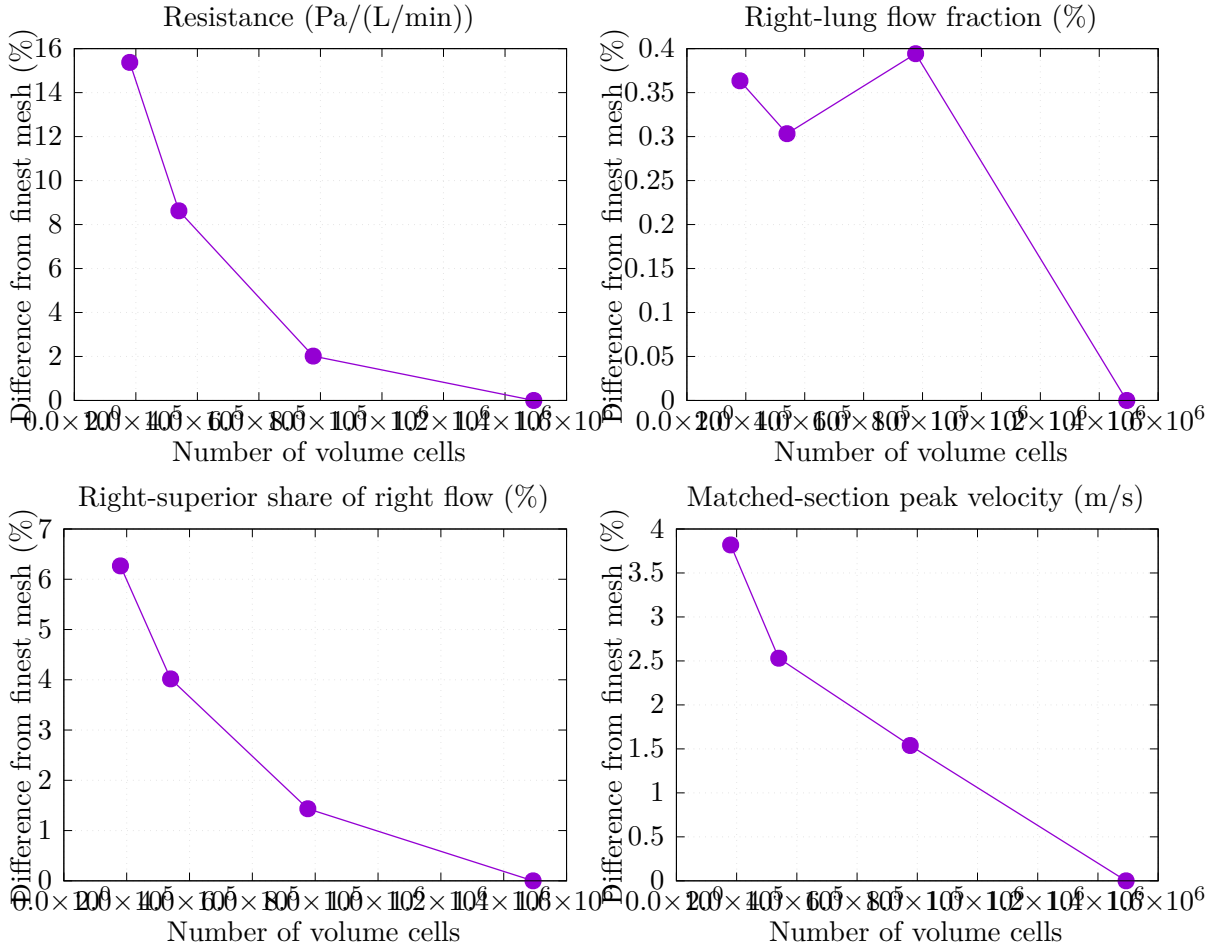


Figure 8: Absolute relative differences from the densest 0.12 mm mesh, calculated as $|\phi_i - \phi_d|/|\phi_d| \times 100\%$. The densest case reached its 2000-iteration limit without satisfying residual control and is therefore a requested comparison rather than an accepted truth solution.

9 Results

9.1 Velocity field

At the converged postoperative solution, velocity vectors accelerate through the narrow distal trachea and divide at the carina. The fixed matched section has an area-averaged axial velocity of 6.53 m/s and a sectional peak magnitude of 9.25 m/s. The vectors also show the unequal right/left flow division quantified from patch fluxes.

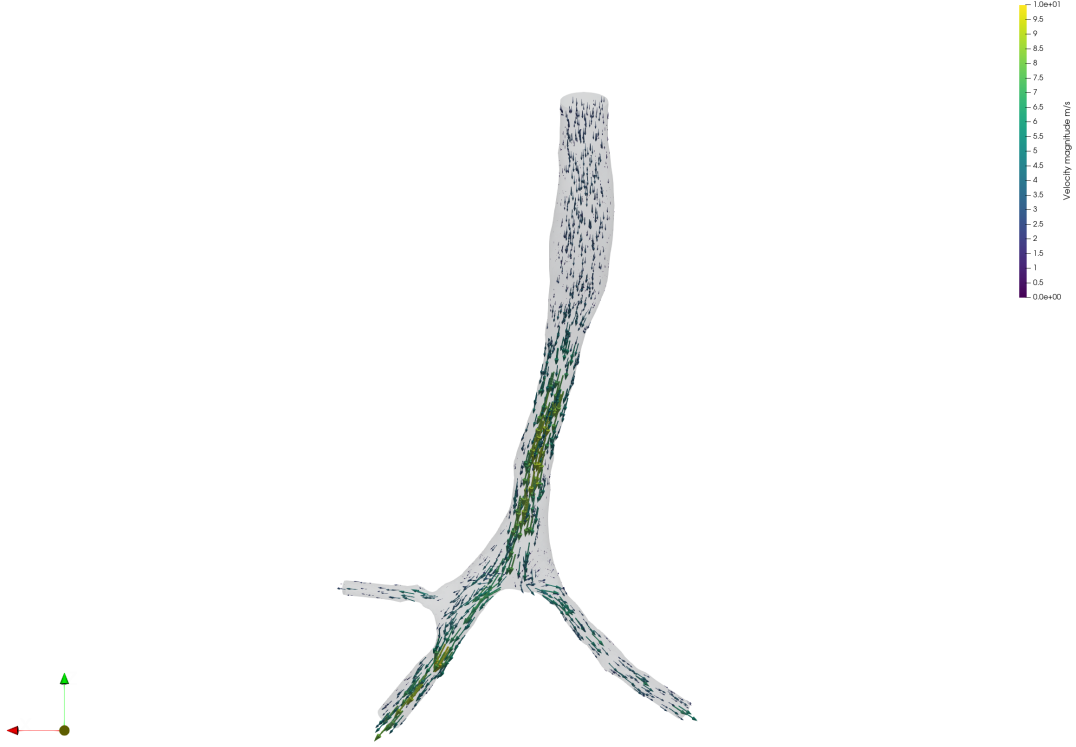


Figure 9: Postoperative velocity-vector field at the converged steady solution (iteration 589). Arrow direction represents local flow direction; arrow colour represents velocity magnitude on a fixed 0 m/s to 10 m/s scale. The airway surface is shown semi-transparently for anatomical context.

9.2 Pressure distribution

Pressure is reported relative to the zero-gauge-pressure outlets and converted from OpenFOAM kinematic pressure using $P = \rho p$ with $\rho = 1.204 \text{ kg/m}^3$. Pressure decreases from the tracheal inlet toward the outlets. Along the mapped centerline interval it falls from approximately 65.1 Pa to 34.1 Pa, with a local minimum followed by partial recovery. The resistance calculation uses area-averaged plane pressures rather than centerline point samples.

9.3 Outlet and lung flow distribution

Outlet flow was obtained by integrating outward-normal volumetric flux over each patch. The right superior and inferior lobar outlets were grouped as the right lung; the left main bronchus formed the left group. The primary left/right result was

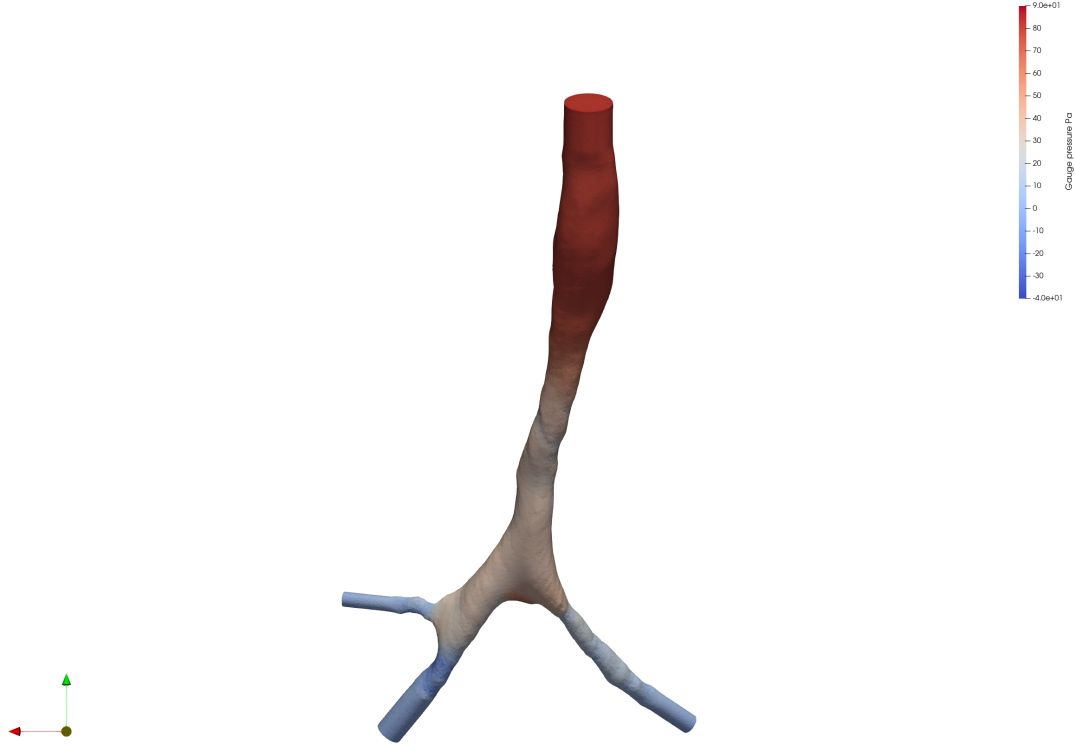


Figure 10: Postoperative dimensional gauge-pressure distribution at the converged steady solution (iteration 589), shown in anatomical frontal view. OpenFOAM kinematic pressure was multiplied by an air density of 1.204 kg/m^3 ; outlet gauge pressure is zero.

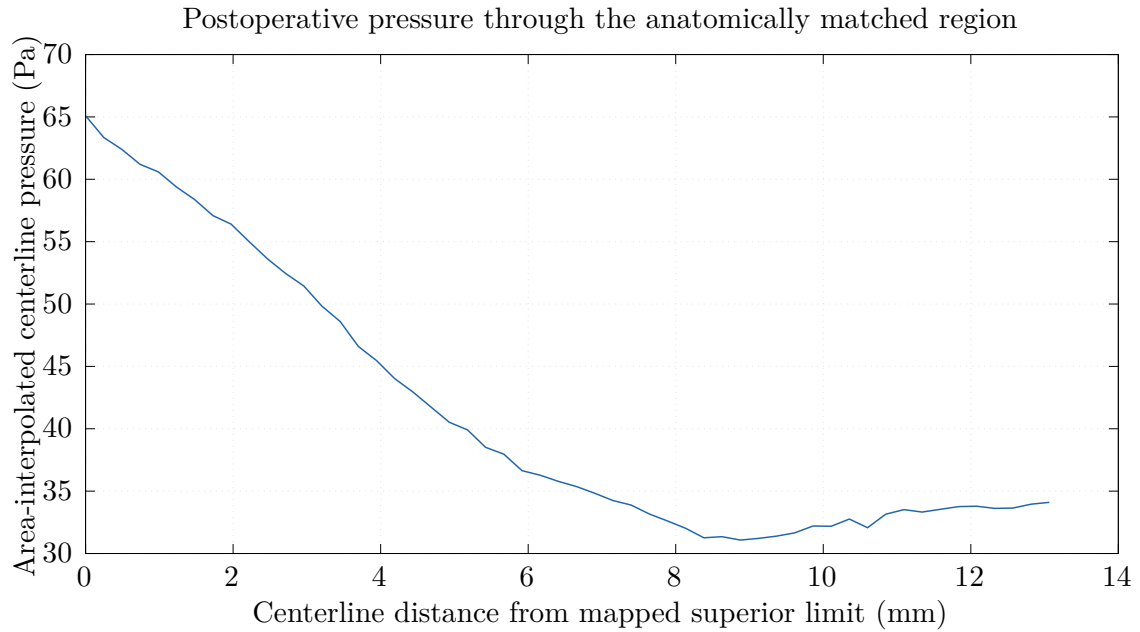


Figure 11: Centerline-interpolated dimensional pressure across the postoperative region corresponding to the preoperative stenosis. Distance is measured from the mapped superior limit toward the mapped inferior limit. This profile shows local pressure evolution; resistance is calculated separately from area-averaged pressures on the bounding planes.

$$Q_R = 73.17\%, \quad Q_L = 26.83\%. \quad (12)$$

The individual outlet values in Table 11 provide the supporting lobar decomposition. Because identical zero gauge pressure was imposed at all outlets, no 73/27 split was prescribed. The unequal distribution instead reflects the geometric and hydraulic resistance of the branches resolved within the truncated airway domain.

Table 11: Outlet flow rates and fractions.

Case	Outlet 1 (%)	Outlet 2 (%)	Outlet 3 (%)	Imbalance (%)
Postoperative	11.23	61.94	26.83	6.0×10^{-7}

Table 12: Postoperative left/right lung flow distribution.

Lung	Flow rate (m ³ /s)	Fraction (%)
Left	8.9449×10^{-6}	26.83
Right	2.4388×10^{-5}	73.17

9.4 Local resistance before the bifurcation

Two planes normal to the centerline delimit a fixed section before the bifurcation, corresponding approximately to the location of the preoperative constriction. These same plane definitions are reused for every mesh and for the transient Assignment 6 analysis. With dimensional pressure difference ΔP and volumetric flow Q , local resistance is

$$R(t) = \frac{\bar{P}_{\text{up}}(t) - \bar{P}_{\text{down}}(t)}{Q(t)}. \quad (13)$$

For the incompressible OpenFOAM pressure field p , $\Delta P = \rho \Delta p$. Area-averaged kinematic pressures were $53.70 \text{ m}^2/\text{s}^2$ upstream and $27.99 \text{ m}^2/\text{s}^2$ downstream. With $\rho = 1.204 \text{ kg/m}^3$, the local pressure drop was 30.95 Pa ; at $Q = 3.347 \times 10^{-5} \text{ m}^3/\text{s}$ this gives $R = 9.25 \times 10^5 \text{ Pa s/m}^3$, equivalent to 15.41 Pa/(L min) . The matched-plane mean axial and peak velocities were 6.53 m/s and 9.25 m/s , respectively.

Table 13: Steady postoperative local-resistance calculation.

Quantity	Value
Upstream area-averaged kinematic pressure	$53.695 \text{ m}^2/\text{s}^2$
Downstream area-averaged kinematic pressure	$27.993 \text{ m}^2/\text{s}^2$
Pressure difference ΔP	30.945 Pa
Volumetric flow Q	$3.347 \times 10^{-5} \text{ m}^3/\text{s}$
Resistance $R = \Delta P/Q$	$9.247 \times 10^5 \text{ Pa s/m}^3$

10 Transient postoperative analysis

10.1 Time-step convergence

The complete cycle comprised 355,016 adaptive timesteps. Of these, 354,399 (99.826%) satisfied the PIMPLE outer criterion and 617 reached the limit of three correctors. The latter were



Figure 12: Fixed centerline-normal planes delimiting the postoperative matched tracheal region. The blue superior plane is the upstream pressure section and the orange inferior plane is the downstream section. Their origins and normals are archived in a versioned JSON definition and reused unchanged for steady, mesh-sensitivity, and transient analyses.

localized around zero-flow transitions: 180 near startup, 309 around the inspiration–expiration reversal at $t = 1$ s, and 128 near cycle end; no failures occurred over the main high-flow intervals. The maximum Courant number was 5.296 and the mean Courant number never exceeded 0.540. The timestep varied from 3.287×10^{-6} s to 2.019×10^{-4} s. Final equation residuals were typically of order 10^{-9} , while the maximum absolute cumulative continuity error was 5.25×10^{-4} . Thus the run remained bounded and predominantly converged, but reversal-adjacent states and precise transient extrema are interpreted cautiously.

10.2 Velocity vectors at representative phases

Three saved states were selected directly from the waveform: $t = 0.24$ s represents accelerating inspiration at approximately 68.5% of peak flow, $t = 0.50$ s is peak inspiration, and $t = 1.50$ s is peak expiration. The two peak states isolate reversal of the velocity field at equal flow magnitude, while the accelerating state shows development of the inspiratory jet.

10.3 Breathing waveform and time-varying resistance

The applied section flow and local pressure difference were evaluated on the same planes as Assignment 4 at all 100 nonzero saved times. Flow ranged from -6.288 L/min to 6.288 L/min, and pressure drop ranged from -58.70 Pa to 167.45 Pa. Resistance was masked where $|Q|$ was below 5% of peak flow because division by a near-zero flow is ill-conditioned. Over the retained states it ranged from 5.71 Pa/(L min) to 26.63 Pa/(L min). Unlike ideal constant Poiseuille resistance, the patient-specific result depends on flow direction and phase due to developing flow, curvature, branching, separation, and unsteady inertia. The preoperative airway was not simulated transiently; its 64.9% area constriction supports only a qualitative expectation of

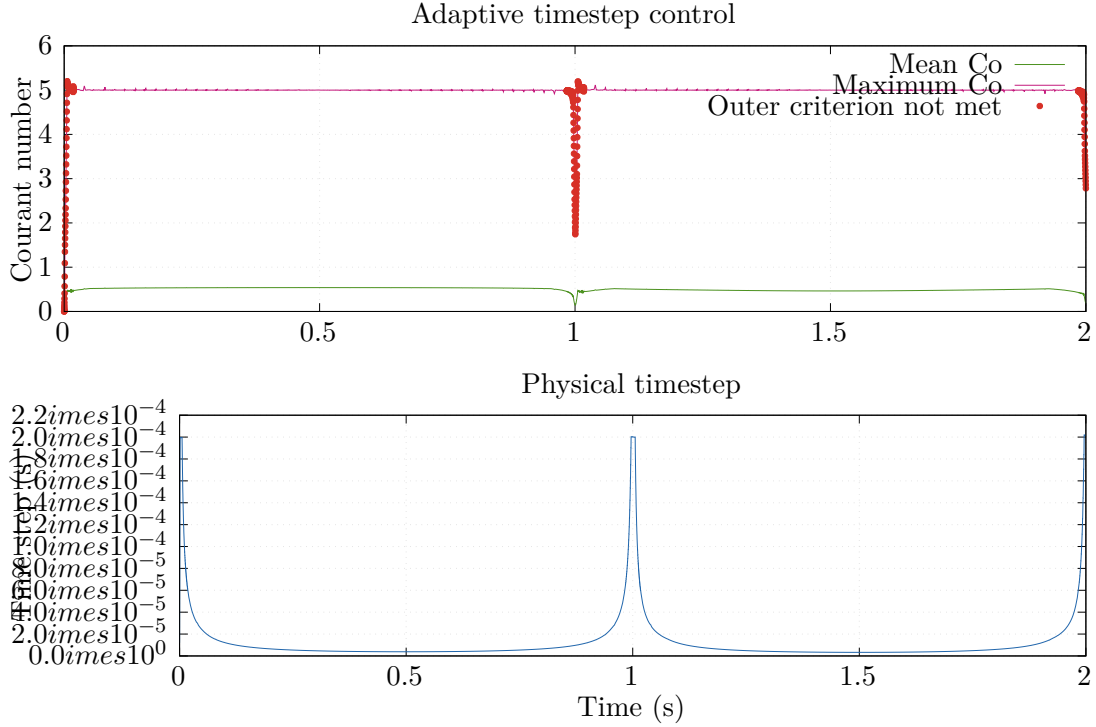


Figure 13: Adaptive timestep behaviour over the postoperative proof-of-concept cycle. Red markers denote sampled timesteps that reached three PIMPLE outer correctors without meeting the selected outer criterion. These events cluster around zero-flow transitions rather than peak-flow intervals.

higher resistance.

11 Discussion

11.1 Interpretation of outlet conditions and lung-flow split

Applying the same static pressure at every distal outlet deliberately leaves the left/right flow division unconstrained, making it an output of the reconstructed geometry rather than an imposed boundary-condition ratio. The predicted 73.17% right and 26.83% left distribution is therefore a defensible measure of the relative aerodynamic resistance of the resolved postoperative branches. Differences in calibre, pathway length, branching angle, curvature, and local flow separation can all contribute to the unequal result.

This distribution should not, however, be interpreted as the child’s measured physiological lung ventilation. The model ends at truncated bronchial outlets and therefore excludes the resistance of unresolved peripheral airways and the compliance of the distal lungs. A more physiologically detailed extension could apply distal resistance conditions such as

$$P_i = P_{\text{alv}} + R_i Q_i, \quad (14)$$

or coupled resistance–compliance outlet models. Such conditions would preserve an emergent flow split while representing peripheral lung mechanics, provided that patient-specific parameters were available.

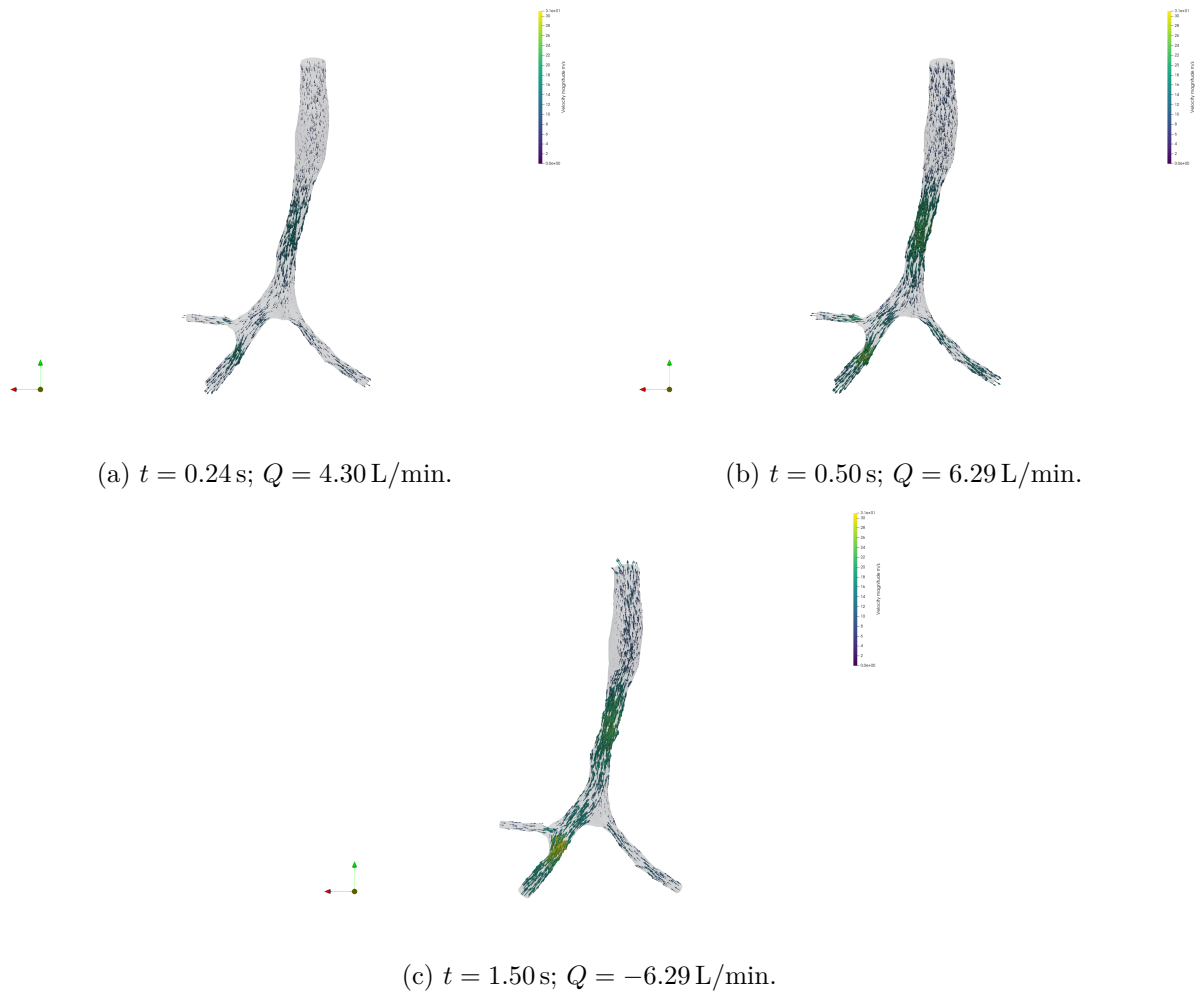


Figure 14: Postoperative transient velocity vectors using one anatomical frontal view and a common 0 m/s to 31 m/s scale. The third panel demonstrates reversal during expiration.

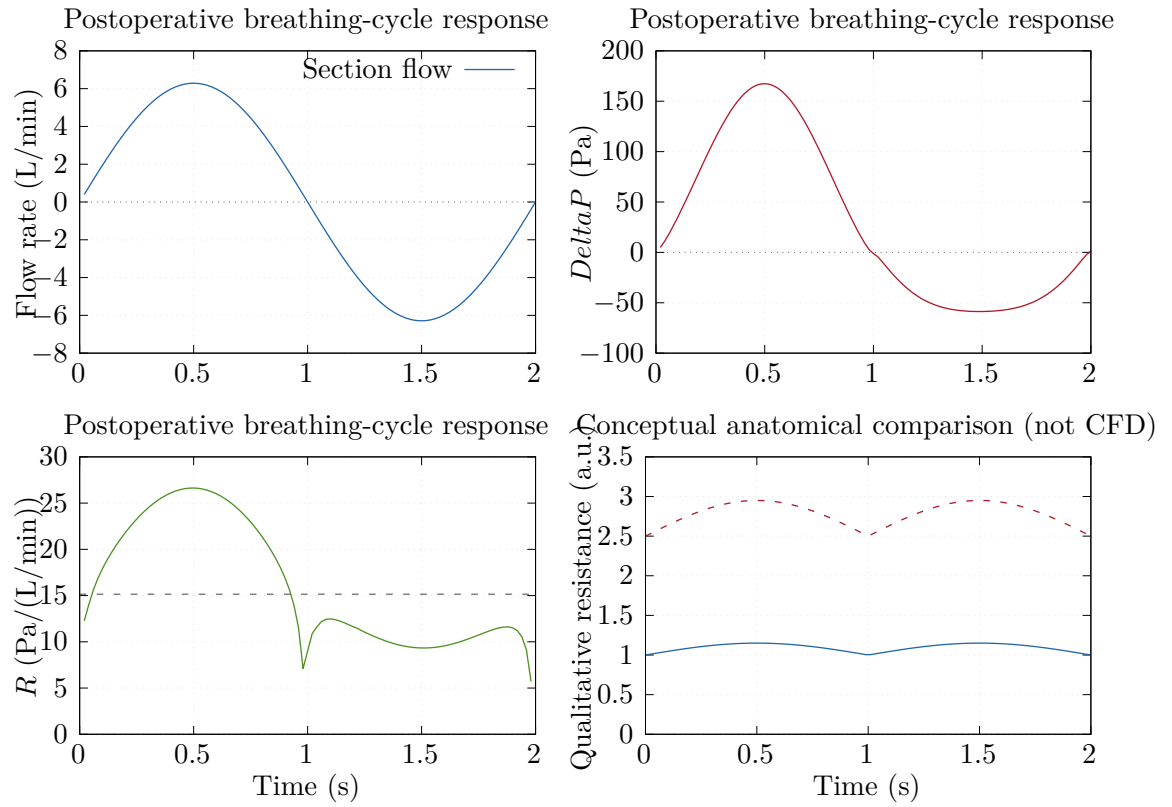


Figure 15: Complete-cycle section flow, local pressure difference, and postoperative local resistance. Resistance is omitted when $|Q| < 0.314$ L/min (5% of peak flow). The dashed line is the steady coarse-mesh resistance at 2 L/min; it is a reference, not an expected constant for this patient-specific unsteady flow. The arbitrary-unit comparison is explicitly conceptual: only the postoperative cycle was simulated transiently.

11.2 Mesh sensitivity and optimal mesh

Resistance and the small right-superior branch split were more mesh-sensitive than total right-lung fraction. From 0.15 to 0.12 mm, all selected metrics changed by at most 2.02%, while cell count increased by 92% and solver execution time by 157%. The 0.12 mm case also failed residual control at 2000 iterations, whereas 0.15 mm converged in 1611. With a practical 2.5% acceptance threshold, the 0.15 mm mesh is selected as the optimal balance between apparent spatial convergence and computational efficiency.

11.3 Constant-diameter resistance during inspiration

For fully developed laminar flow in a rigid constant-diameter tube, Poiseuille’s law gives $\Delta P = 128\mu LQ/(\pi D^4) = RQ$. The viscous pressure loss therefore increases linearly with flow, but $R = \Delta P/Q$ remains constant. Thus, it would be imprecise to attribute the observed increase in effective resistance to linear viscous friction alone.

A useful reduced description of the patient-specific transient pressure loss is

$$\Delta P(t) = aQ(t) + b|Q(t)|Q(t) + L_f \frac{dQ}{dt}, \quad (15)$$

where aQ represents approximately linear viscous loss, $b|Q|Q$ represents flow-dependent convective and form losses, and $L_f dQ/dt$ represents fluid inertance. Dividing by Q away from flow reversal gives an apparent resistance containing $a + b|Q|$ and an acceleration-dependent term. Consequently, effective resistance can rise with inspiratory flow as jets, developing profiles, curvature, branching, and separation increase dissipation. This is consistent with the computed inspiratory increase from approximately 12.3 Pa/(L min) at 0.40 L/min to 26.6 Pa/(L min) near peak inspiration. The lower and asymmetric expiratory branch is physically possible for reversed flow through a non-symmetric geometry, but its magnitude remains sensitive to inertia, outlet conditions, the initial cycle, and the coarse high-Courant discretisation.

11.4 Limitations

Potential limitations include:

- static CT geometry despite the dynamic nature of tracheomalacia;
- rigid airway walls and absence of fluid–structure interaction;
- an idealised sinusoidal waveform rather than a measured patient-specific waveform;
- idealised outlet pressure conditions;
- uncertainty from segmentation and clipping-point placement;
- an isotropic first-order tetrahedral mesh without dedicated prism boundary layers;
- the selected laminar-flow assumption; and
- analysis of a limited number of anatomical cases.

With additional resources, refinement should focus locally on the matched tracheal region, carina, and small right-superior branch rather than reducing global size uniformly. All cases should first meet residual and monitored-integral criteria. Additional levels around the apparent asymptotic range, consistently generated prism layers, Richardson extrapolation, and a grid-convergence index would strengthen numerical uncertainty estimates. Runtime, memory, mesh quality, and section-area variation should be tracked alongside each physical metric.

11.5 Handling transient non-convergence

The 617 outer-criterion failures were concentrated around startup and the two zero-flow transitions, where pressure and velocity fields must adjust while the flow changes direction. The first remedy would be to repeat those intervals with a lower Courant limit, because this directly improves temporal resolution and pressure–velocity coupling. If failures persisted, the number of outer or pressure correctors would be increased. A converged previous cycle would provide a better initial condition than rest, and additional cycles would be simulated until corresponding phase metrics became periodic. Finally, the cells identified by the three failed `checkMesh` checks should be repaired before a quantitative production run. These changes target the observed causes without misrepresenting the present proof-of-concept result.

12 Conclusions

The preoperative minimum lumen area was 1.813 mm^2 , compared with 5.169 mm^2 at the corresponding postoperative location, giving a 64.9% area constriction and demonstrating a substantial anatomical increase after intervention. Steady postoperative CFD at 2 L/min predicted a local pressure drop of 30.95 Pa, resistance of 15.41 Pa/(L min), peak velocity of 9.25 m/s, and a geometry-driven 73.17/26.83% right/left flow split. The 0.15 mm HXT mesh was selected for steady analysis because its monitored metrics differed from the densest mesh by at most 2.02% while avoiding its convergence and cost penalty.

A complete postoperative transient proof of concept produced the prescribed $\pm 6.283 \text{ L/min}$ cycle and direction-dependent resistance between 5.71 Pa/(L min) and 26.63 Pa/(L min) outside masked zero-flow intervals. Although 99.826% of timesteps met the PIMPLE criterion, the coarse mesh, $\text{Co}_{\max} = 5$, failed mesh-quality checks, and absence of timestep independence limit this result to qualitative phase trends. The anatomical and steady integral conclusions are therefore stronger than the transient extrema.

Reproducibility statement

The complete workflow is documented in `WORKFLOW.md`. Record the Git commit, Slicer version, Gmsh version, OpenFOAM image identifier, mesh sizes, cell counts, and solver settings used for the final results. The scripts `create_volume_mesh.sh`, `run_cfd.sh`, `run_fine_cfd.sh`, and `fetch_cfd_results.sh` automate the meshing, remote execution, reconstruction, and retrieval stages.

A OpenFOAM

OpenFOAM is an open-source computational continuum-mechanics framework that uses the finite-volume method. Conservation equations are integrated over each mesh cell and the divergence theorem converts volume derivatives into fluxes through its faces. Interpolation and gradient schemes provide face values from cell-centred quantities, producing sparse algebraic systems for velocity $\mathbf{U}$ and pressure p [3]. OpenFOAM uses kinematic pressure for incompressible flow,

$$p = \frac{P}{\rho}, \quad (16)$$

with units of m^2/s^2 ; dimensional pressure differences in this report are therefore recovered from $\Delta P = \rho \Delta p$. For incompressible Newtonian laminar flow, the solvers considered here discretise

$$\nabla \cdot \mathbf{U} = 0, \quad (17)$$

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{U}\mathbf{U}) = -\nabla p + \nabla \cdot \left[\nu \left(\nabla \mathbf{U} + \nabla \mathbf{U}^T \right) \right]. \quad (18)$$

The transient term is omitted in a steady calculation. Pressure and velocity are coupled physically but solved in a segregated manner. A discretised momentum equation may be written schematically as

$$A_U \mathbf{U} = \mathbf{H}(\mathbf{U}) - \nabla p. \quad (19)$$

Using the reciprocal diagonal coefficient $r_{AU} = 1/A_U$ gives a predicted velocity and face flux. Requiring the corrected flux to be divergence-free then produces a pressure equation of the form

$$\nabla \cdot (r_{AU} \nabla p) = \nabla \cdot \phi_{H/A}. \quad (20)$$

The resulting pressure correction updates both velocity and face flux, thereby progressively enforcing mass conservation. On a non-orthogonal mesh, pressure correction is repeated so that explicitly treated non-orthogonal contributions are updated before the final conservative flux correction.

A.1 The `simpleFoam` solver

`simpleFoam` is a steady-state incompressible solver. It uses the SIMPLE algorithm: (1) assemble and solve the momentum equations using the current pressure; (2) construct and solve a pressure-correction equation; (3) correct pressure, velocity, and face flux; (4) update boundary conditions; and (5) repeat until convergence. Its iterations are numerical fixed-point iterations and do not represent physical time. Under-relaxation limits field changes between iterations and can stabilise pressure-velocity coupling.

In this study, pressure is solved using GAMG and velocity with `smoothSolver` using symmetric Gauss-Seidel smoothing. Two non-orthogonal correctors are applied in the steady case. The SIMPLE residual controls are 10^{-5} for pressure and 10^{-6} for velocity. The Assignment 4 postoperative solution met these criteria after 589 iterations. Convergence is additionally assessed through integrated inlet/outlet flow because small algebraic residuals alone do not guarantee conservation of the reported quantity.

A.2 The `pimpleFoam` solver

`pimpleFoam` advances the incompressible equations through physical time using PIMPLE, which combines SIMPLE-like outer iterations with PISO-like inner pressure correctors. At each new time level, time-dependent boundary conditions are updated and a momentum equation is assembled. Each outer corrector updates the nonlinear momentum terms; within it, pressure correctors repeatedly enforce continuity and update p , $\mathbf{U}$, and the face flux. More correctors can improve coupling at a timestep but increase its computational cost.

The selected-mesh transient configuration uses the second-order backward time scheme, at most three outer correctors, two pressure correctors, and one non-orthogonal corrector. Its timestep is adjusted to limit the maximum Courant number,

$$\text{Co} = \frac{|\mathbf{U}| \Delta t}{\Delta x}, \quad (21)$$

where a smaller limit improves temporal resolution but requires more steps. The coarse proof-of-concept configuration instead uses bounded upwind convection and two non-orthogonal correctors; its Courant limit is recorded with the final results because it was varied during timing tests.

An OpenFOAM linear-solver *initial residual* is the normalised algebraic defect before an individual equation solve, whereas its final residual is the defect after that solve. These residuals do not directly measure discretisation error or physiological accuracy. PIMPLE outer residual controls assess whether the coupled fields reach their per-timestep targets, here 2×10^{-3} for velocity and 10^{-2} for pressure. If these targets are not met within three outer correctors, `pimpleFoam` prints `PIMPLE: not converged within 3 iterations`; it nevertheless retains the latest corrected fields and advances. Such a timestep therefore has a computed solution, but did not satisfy the selected nonlinear stopping criterion. The frequency of these events, linear residuals, continuity errors, Courant number, and monitored flow quantities must all be considered when judging the result.

B Parallel computation with the Message Passing Interface

The Message Passing Interface (MPI) is a standard for communication between independent processes, called ranks. Each rank has separate memory and performs local calculations while exchanging explicitly required data with other ranks. OpenFOAM couples MPI to spatial domain decomposition: `decomposePar` reads `decomposeParDict` and partitions the finite-volume mesh into one subdomain per rank. The Scotch method used here partitions the mesh graph automatically, seeking a similar cell count per rank while limiting the number of faces cut between subdomains.

A cut internal face becomes a pair of coupled `processor` patches on the two adjacent ranks. These are numerical interfaces, not physical airway boundaries. Before interpolation, gradient evaluation, or sparse matrix multiplication requires a neighbouring value, ranks exchange interface or “halo” data. Fluxes across processor patches are coupled so that the parallel calculation preserves the same conservation law as the undecomposed mesh.

Some operations require collective communication across every rank. Global sums are used for residual normalisation, continuity errors, convergence tests, and integrated fluxes. GAMG also communicates during restriction to coarser levels, coarse-grid solution, and prolongation. Communication and synchronisation become relatively expensive when each rank owns too few cells; therefore doubling the number of ranks does not generally halve runtime. A good partition balances cell counts while minimising processor interfaces, but the most efficient rank count must ultimately be established by timing a representative calculation. This consideration motivated testing fewer ranks for the approximately 180,000-cell coarse transient proof of concept.

During a parallel run, each rank writes its fields beneath `processor0`, `processor1`, and subsequent processor directories. After simulation, `reconstructPar` maps these distributed fields back to the original undecomposed mesh for serial post-processing and ParaView. Reconstruction changes the storage layout, not the numerical solution. The project scripts consequently decompose before execution, launch `simpleFoam` or `pimpleFoam` through `mpirun`, `reconstruct` the retained time directories, and omit bulky processor directories when fetching results.

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